

I would hate to see a future where teachers outsource to AI the parts that I think really make good education, which is the connection pieces, when you really understand your students and can spend time with them, and AI can be used in so many ways that allow teachers to have more time to do that kind of work. And I'm excited for us to talk with institutions and discuss with them ways where we can amplify that knowledge they already have. - Hi everyone. We're here to talk about my favorite topic, which is AI and education. My name is Drew Bent. I lead our work here in education on beneficial deployments. Formerly was a high school math teacher. My parents are educators. I worked in education nonprofits and would definitely consider myself a lifelong learner. I'm joined here by my wonderful colleagues who work in education across the organization. Do you wanna start, Zoe? - Happy to. Yeah. Hi, I'm Zoe. I'm on our education team here at Anthropic and I support all of our non-technical audiences, including educating teachers and students about both our products and AI in general. - Hi, I'm Maggie and I founded and currently manage and support said education team, which we fondly internally call the Ministry of Education. - Hi, I'm Ephraim. I'm a product engineering manager and I've also helped build some of our products for facing education. - So I think it's helpful to start here with why are we discussing education in the first place or why do we even work on education in the first place at this general purpose AI lab? You know, we all know of course that at Anthropic we care a lot about studying both the potential of the technology that we're building, but also the risks. I think education is the perfect example and sort of embodiment of that because of course there's massive benefits as we'll talk about in this conversation. But there's also a lot of concerns we have about the impacts of AI and education. And so when we think about the benefits, we think about, you know, I've had many conversations with you all about how AI can prevent teacher burnout, how it can transform and really democratize access to high quality learning and tutoring, how it can change, you know, how and what teachers teach. But then we also of course see the other side of it, which are all the risks and the concerns around, you know, that teachers have about how AI could lead to more cheating and is leading to more cheating of course, but also serve the more existential risks of how do we make sure that these tools are actually enhancing and augmenting human thought as opposed to replacing it. And so my hope for this conversation is that we can sort of dig into all of these nuances but also talk about the, the sort of practical type of work that we're doing at Anthropic to work towards these issues. So maybe to get started, I would love to hear, you know, I know all of you, I've known you for a while, but I don't necessarily know all of your

stories on what got you interested working on education in the first place. So Maggie would love to start with you and what brought you into this work?

- Well, my interest in education is twofold. I think professionally, education and communication has always been part of every job I've ever held up until Anthropic. And I think personally I, you know, have two lovely kids in my life and I am grappling with just

as every other parent is in this era, what can I do to help them to being kind of intelligent, thoughtful,

like thinkers and and critically engaged individuals as they grow up in the AI age.

One is a lot more professional interest and where I feel like I can make a difference and a latter is like pressingly concerning to my immediate like core as someone who is a guardian to young minds. - How about you Ephraim? - Well, so I started my career in academia.

I studied physics, maths, and I assumed I was going to be doing research for most of my life before switching to tech.

And I, I've taught classes when I was at MIT, I've been in adjunct faculty, so education has been something I've always been interested in.

When it comes to AI and education, I have two children who are in college, so I worry every day about what they're learning, how they're learning, what will they do once they graduate. I'm also very passionate about, actually we'll talk about later on, how do institutions deal with AI in, in their, in their education.

So just a lot of stuff here that is both personal but also like looking forward to society. What is it, you know, what does AI mean to education that I'm really interested in. - I mean I think like you and I both having children in our lives is like one of the most pressingly concerning things.

Yeah. Where it makes it so real for you right now. Right, right. And I think that like your kids are college age

and so they're trying to figure out what they're doing in their lives and then my kids are younger, but I see it on the horizon

that very soon there's like strong decision points where you can decide how to start nurturing this kind of thinking.

And I feel like if you don't catch it early, which is why we care about like education at all the different ages that it can kinda go pretty

badly and compound. Right. - Yeah, I feel like that's hitting on a lot of like why I got into education in the first place,

which is I have this very deep seated belief that education is one of the most important things we can do to make change in our society.

I don't, I think most people can agree on that and when I, you know, pivoted from the classroom into tech, it was because I wanted to work for organizations

that could do that change at scale. I think there's some things with our education system that I would fix if I had a magic wand

and I, I am hoping that AI can help accelerate some of that change for the better, but also like very much recognizing

that we have big responsibility to make sure that that change goes well today. And

then, you know, 10 years into the future. - There was a great quote from a professor that I talked to at some point where they were saying that all the problems in academia have existed

for a while as an institution. It's just that AI is the forcing function that makes everyone deal with it now instead of keep putting it down, I guess kicking the can down the road, right? - Yeah. - Yeah. So I'm excited for us to deal with that ourselves.

- I'm just realizing now we have the parents on this side, we bring a good set of perspectives here and yeah, I think on my end also my parents are, are educators and so it's always sort of looking up to them and wanting to do what they did. But I think it's helpful to also ground this conversation in some of the research that we've all been working on here. And so it was I think late last year that our Societal Impacts team at Anthropic had done research into all the ways that, you know, users are using Claude and found that some of the top uses were in education.

I think we had seen this sort of in all chatbots to some extent, but I think it was also a wake up call for us because you know, what's interesting about it is that these LLMs, as we all know, were built not with education in mind very much on answering questions. They're fine tuned that way. They're meant for, you know, these productivity tasks, right? And then it's kind of this interesting emergent, you know, phenomenon that they're very helpful for education potentially sometimes destructive as well for people's learning.

And so we start to, you know, dig, dig deeper into that. But I think what stood out in the research is, one stat that's always comes up is 47% of the student interactions on Claude were very direct transactional types of interactions with little engagements. And I think, I know when Maggie and I were looking at data first is sort of sort of a wake up call because again we, we have all these incredible ways that you can use it as a Socratic tutor, but then to see that in some cases, you know, people are using it to just do their homework.

And I know as a teacher, like for me, I sort of think about the different cognitive skills that I want, you know, my students to learn and you know, at the base level and maybe things like remembering a fact and understanding some knowledge, but then you wanna eventually get them to the level of synthesis and you know, creation of course, you know, we call this Blooms taxonomy, but what we saw in the data that I think was fascinating was we start to study Claude's interactions in these conversations and saw how well Claude is performing on these cognitive tasks and found that Claude was performing at the these top levels of creating and analyzing when, again, as a teacher that's what you want your students to do. - Yeah, I think the students are kind of flipping the script on this in a way that's concerning to us as educators. And I don't know if that's necessarily, I think the first blush reaction is that that's a bad thing. But I think part of what I want to challenge us to think about and and the world to

think about is, is there like a novel taxonomy where that's the baseline and then you can build on top of that to something new that hasn't been possible before AI, - We've also explored how the, you know, educators are using it and they're experimenting with it for, you know, create lesson plans, grading.

I think it was one Northeastern professor who told us that they're never going to assign a traditional essay again because they just had too many students submitting, you know, these AI assignments.

Whether they use Claude or not, we don't know. But I think that's that sort of raises a lot of questions for us.

- I feel like you really hit on two of the things that we talk about a lot with educators, which is like AI is both changing how students learn in, in like also what they need to learn, right? Like I don't actually know if it's important that students have the same memorization

that they would've needed to have like 10 years ago because they have AI tools readily available or in theory they should have AI tools readily available.

And then, you know, when you get into like higher levels of academia, there are potentially skills that we're teaching today that won't be

as important in the future. And so it's, it's a ton for teachers to grapple with. - I would love to hear from all of you,

what is one thing you're really excited about with how AI can transform teaching and learning? - I think one thing that really stands out to me is

interactive learning experiences. I have this very vivid memory of when I was in the classroom my students did a virus

simulator game that was fully programmed. They were the virus and they worked into the cell and they replicated.

And the engagement that I saw from my classroom that day was unlike anything else. And I think most teachers have seen something like this,

but AI really lets you do this at scale with any subject, right? Like imagine you're talking to a historical figure and,

and teachers can put a lot of guardrails around this with the right tools, but I just am very excited to see that space developing over time.

- I think the interactivity is also really interesting to me. There's so much assistance you can get from AI that

is really hard resourcing wise to get the interactivity, especially in like low resource regions where you know,

many students don't have access to like a personal career coach that can walk you through how to interview properly right at an organization

and with the power of an AI like Claude, you can like upload like the job listing your resume and so on

and just ask Claude to help you role play through these things. I think there's a lot of really engaging, interesting role play experiences whether

or not with a dead historical figure or with some sort of like coaching situation

that can really help you through a lot of experiences where an external perspective would be immense help. It's just really hard to get that other human being

to find time to sit down with you, especially in regions with low resourcing. - Related to that I've, I've been very excited about

how teachers are transforming their assessments with it. I was talking to a teacher a

few weeks ago who at one point I think during the pandemic had taken time over, you know, Zoom to basically do oral interviews with all the students and really get to assess them on a more holistic way. But of course that didn't scale very well and so stopped doing it. But then with these AI tools coming out was able to now sort of use the same rubric and start to have, you know, all the students doing on a regular basis these sort of assessments back and forth with the chat bot and then the professor, the teacher is able to review them and assess them based on that process of going back and forth with an AI. - I think assessment is a very interesting use of AI. I could, you know, envision in the future where it isn't a specific moment in time where you get assessed but there is a continuous interaction with AI developing a much deeper understanding of do you really understand the concept behind this algebra concept or not. One thing that I'm really excited about AI could provide this personalized learning. There is a research done on one-on-one tutoring and what they found is that on average the average student that had one-on-one tutoring is better than the 98th percentile of students that didn't have one-on-one tutoring or just a classroom set up. - And that was with human tutoring. - That's with humans, right? Hard to scale. Exactly hard to scale. Assume maybe like an hour of one-on-one tutoring a day. With AI you get continuous one-on-one tutoring and that is available to everybody, you know, around the world. So I think that is, has a great potential to transform the world and how people learn. Yeah, I agree. I think there's of course lots of challenges that as people have looked into the studies and how do you replicate this and all of it. But I think it's a very useful sort of north star of what could be possible if you can have a very personalized but also personal type of tutoring experience. - Absolutely. I think today we have classes maybe segregated by students that are in the top or you know, you take a like an AP class if you are, you know, in that certain group or not. But with AI every student could have their own journey. Those that are able to advance could advance very quickly and those that need help can get that personalized help. - You know, this reminds me of a really interesting use case that I talked to a teacher about where, you know, you kind of always wanna meet students where they're most interested, right? Like that's like a very Montessori type of approach where it's like what is your favorite topic and then we'll kind of match all the subjects to that. That's really hard to scale, right? But there's a teacher I talked to that's just like, I ask my students what their favorite things are, they tell me a little story and then now every single handout that you have, like the same math concepts, maybe even same problems, but it's like each handout is made for each student and it's like exactly according to their interests.

It's got a story that's engaging to them, problems they actually care about. And like she's noticed an uptick in the engagement for sure because suddenly these students have a through line, through every subject in the classroom is building on itself in a way that's super personalized to their interests. Which like if that was in every classroom, imagine how much students would just lean into, it's amazing. Like exactly. - Yeah. So how are you thinking about that question of what's worth, you know, learning in, in the age of AI

- As somebody who's in product, in product development, what I see is this absence of product layer that would help both students and teachers use AI very effectively. For example, my daughter's class, she's taking Python, so both of my children are studying computer science, so very, very relevant for their exams for writing Python. They want them to write on a piece of paper because they're afraid about cheating. Well the reason it is so challenging now is there aren't products for the students could use to learn. There's also not products for the teachers to use to assign and grade homeworks, all of this are very light lifts, like from what we can do as a, as a product offering. But in the absence of an intentional product that is built on top of LLMs exposes a lot of uncertainty, fear, and abuse of the technology that we're building. So that's, I guess that's my take on this is that, you know, with a little bit of support in product thinking, so much of the, the sort of this uncertainty and and and maybe like cheating and so on could be mitigated. - And one of the things I struggle with is, you know, we can sort of backtrack from how jobs are changing and start to think about how university education should be changing. But then when we think about, you know, kids to the, the age of your kids, you know in K 12 it's an even harder question about what will be the skills, the durable skills that they need years from now. So I don't have the answer, but look, I always look, I look to you Maggie. - Oh man, it's, it's a challenging one. I think that what resonates a lot with the teachers that I talk to is that a lot of the skills you teach young minds about how to critically think about the world around them in the world of humans can be quite applied to AI. Especially in the world of just critical thinking about the facts that you're presented with. There's a stage of development where you go from trusting every single thing everyone says to you to starting to think about well what are the other things that you need to know in order to believe that this is true with my kids? It's like a two part framework where part one is just the importance of education I think is more important than ever where you can't tell if an AI is bad at math, if you are bad at math or you don't actually know what the right answer is, right? And we're not the stage where AI is like always reliable

like a calculator or something. And so just understanding that and emphasizing that learning, reading, writing, science, math and so on is still very important. And the latter part is developing them into like critical

consumers of information where it's not just about this is what a fact that's given to me is,

but like why is that the case? How do I trust that that's true? What are the areas I need to check in order to ensure

that I can kind of corroborate what I'm learning here? And that kind of critical thinking skill you can develop from a pretty young age regardless of if it's an AI giving you the information or another human being giving you the information. That kind of critical thinking I think is one

of the most important things to get at a, like an early age that skepticism, curiosity and combination. Right?

- Yeah, I wanna add on to that 'cause I feel like a lot of the like teachers and parents that I talk to feel this like really intense

pressure to have the answers and to know what to teach their kids and to know like how to conduct the lessons in their classroom.

And I think like kids are so much smarter than we give them credit for. And so there's something really profound about just sitting

with whether it's your students or your actual children and just like learning with them, asking AI something

and then evaluating what comes out of that together and like having kids reflect and building their own frameworks for interacting with AI.

That I think is really, really powerful. And we all hear, we don't have the answers, therefore no one does.

Obviously we're working really hard to figure out what they are. But I think just encouraging that reflection at any age

where wherever it's developmentally appropriate - Right, - Is like one of the best things people can be doing right now. - You know, I, I recommend sitting down with

your kids

and going through AI together, right? Like ask a question and then say, well this is was really confidently said,

but is that enough When someone says something confidently, is that enough for you to believe that? And hopefully the answer is no. Right?

What else can you check? Can you look somewhere else? What information do you need to think about this? And genuinely internalize if that's correct or not.

That exercise is, I think super and I think of the, the converse side is to kind of demonstrate what it's like to not know something. I think that a lot of times like showing uncertainty

and modeling for your kids when you don't know something, what is your own process of finding that out, right?

What is your process for learning? I think that like what, what I wanna impart upon my kids is like finding the answer

is just the start of your learning journey. And I think for many institutions, schools and so on, getting to the answer is really

what we're testing right now. But if we get make that the beginning of someone's journey, especially learning with AI, then

that I think opens up a whole series of doorways. And so yeah at home I, you know,

try to show my process

for discovering something and that adults don't always have the answers. Kids are super smart

and they're able to find answers on their own in ways that if we just talk and ask the right questions, then they'll be able to discern for themselves what is true

and what isn't and not just trust things at face value. - Yeah and I love the way you framed that of like modeling

how to work through that problem. Modeling an uncertainty is just such an important thing. Oh yeah. That we're all

uncertain so let's use that to our advantage. - And we don't do that nearly enough I think. Like we wanna project this kind of confidence

and I think sometimes it can be detrimental to development of a child to kind of just tell 'em

to trust everything everyone says or the adults around them always know what they're talking about. 'cause I think that that gives them, it's a crutch, right?

To not actually have to think through like truth for yourself and define your own truth.

- I think one thing that sort of remains unchanged is that how human things learn, right? We, we learn basic things first.

We learn addition, subtraction and kind of keeps building up. So that will remain true whether we have AI today,

generation AI or next generation AI. So I think where I see I, I don't know what is the right field to study is,

but either way we still have to go through this process of learning. And I think the great promise now is that you could actually use AI to advance your learning to,

you know, get gain greater even understanding sort of answer maybe in two different ways. In one way sort of personal is I studied physics

because I, I was very curious as a child, so if you are a curious person, I want to learn about the world.

My god, what a, you know, what an opportunity now. 'cause AI could just teach you about anything you want to know. Yeah. But if you're thinking about sort of career, like

what happens next? You know, how do I make a living then no matter what, we'd have to be able to use the AI technology

to make yourself like you plus AI be a more capable employee.

- I agree. Although I do think some of the, the fundamentals are changing in terms of the order in which we learn things.

So one example I go back to as you know, you talked about programming and what computer science, you know, when I learned how

to program, probably similar with you, I spent 90% of my CS education learning how to write code

and write to algorithms and then maybe 10% of my time learning how to read other people's code and review it.

And then now of course at Anthropic when I program with all the coding agents and Claude code and all of that, I spend, you know, maybe 10%

of my time writing the code but 90% of my time reading the code. And so it has made me wonder, you know, we usually learn how

to read before we write just as kids, - Right? - But then with coding we often spend a lot more

of a time writing and then reading. And so it is starting to make me wonder like do we have to revisit some of those fundamentals and like maybe a core part of an intro is CS students education should be to think about reading and being able to discern good code from bad code and all of these things. So I do want us to bring us back to

what is Anthropic doing here. I think it's important, and we all know of course that we have a responsibility here.

We are building this technology that's having this impact, you know, on the education system even though it, we didn't intend it that way initially. And so we have a responsibility as a company, as a public benefit corporation,

but particularly as individuals working in this company, former educators. And so I think it'd be helpful to sort of talk through

what are the things we're doing, what are we wrestling with? I don't know if you know Zoe or Maggie, you want to talk about some

of the work we've been doing with AI fluency? - Yeah, yeah. Happy to start. So I work on education content.

So that's one of the main ways that I can make a difference in this space. So one of the things I'm really excited about is our AI fluency courses.

We partnered with two professors, Joe Feller and Rick Dakan, who built this really great framework about

how to think about using AI. And what's cool about this is we're taking a step back from the products that are available today and the, the prompting

and kind of like all these hacks that you see out in the - World, there's a lot of them

- And there is a lot of them, right? It's like, it's pretty overwhelming.

- And they get outdated so fast. - They get outdated so fast. And so the idea here is we wanna give people a tool

that they can use to understand the interactions that they're having with AI and work towards interactions that are efficient,

effective, ethical, and safe. That's the AI fluency definition. So we have this core course that I think is pretty great

and then we've also created spinoff courses for educators and for students as well as a longer course for educators

who are interested in teaching AI fluency. And so the idea is that anyone who goes through one of these courses is kind of better equipped

to assess their own AI interactions. I talked about like learning with your students earlier and the power of reflecting on your AI interactions.

And at its core that's really what this course is about. It's just reminding everyone, teachers, students, parents,

that they have autonomy in their AI interactions. So that's one thing I'm excited about. - Yeah, I think the interesting thing about our AI fluency

work is the fact that we're taking a step back to the fundamentals I, when we started this AI fluency work,

but way back when, I don't know if you remember Drew, the question that we were trying to answer was, you know, all of these prompt engineering tips

and so on, they are developed by other humans, right? And it's not as if like we at Anthropic have greater superpowers than everyone else.

Externally, we just have a different way of thinking about approaching models and it's like how do you teach that mindset?

- Yeah. - To somebody. 'cause to Zoe's point, it's like, in all of us, that sounds so cheesy, but like we have that capability to be critical thinkers that engage with this. And I think sometimes the fear of needing to get it right kind of supersedes our ability to just experiment. And what I love about AI fluency is we're opening the door to experimentation and saying you can try these things. They may not work for you and it's just as important to learn when they don't work for you and when you shouldn't use AI as it is to, you know, learn when you can use it. I think there's something that we say in the education team every now and then, which I think resonates a lot, which is like we would much rather teach a million people to not use AI than like watch a billion people to become dependent on the technology, right? And in practice that can be quite hard. But AI fluency I think is a very solid start.

- I still remember when I first heard you say that and I was so happy and I knew I'd come to the right company because here I was at an AI lab and Maggie was saying, yeah, I don't think we should use AI in this case or let's teach people how not to use AI. - It's like giving 'em the tools to make the decision on their own right

- Back to critical thinking every time. - But I think, so of course part of it is an education and a training and awareness part, but we are also building products and models that are used out there in the wild. And so I think the work that Ephraim you and your team have been doing on learning mode is a really important part of it. So we'd love for you to share more about how did they even come about. - So learning mode is a set of features that positions Claude as a tutor to students. Students could come in and for example, and upload their assignments and rather than answer the questions explicitly, it would help the students through the material that is covered in their classroom. It will guide them through how to answer the question. It will tutor them, it will also help them prepare for exams for example, by showing them flashcards based on the content that they've uploaded. It's actually very much a grassroots effort. So a lot of people at the company that are really passionate about education and wanting to add education tools within the main product line. So with learning mode what we did is added, it's really is like small features here and there, but then tailor Claude app to be really good at helping students with learning. Along the line we've also added a few more features like, you know, expanding how much content you could add into projects so that more and more content can go in there, connecting to classroom management systems so that content could flow in and out very easily. So that's just the starting point and of like, you know what I think this could be in the future. - And I think what's interesting is some of that early research that led into to learning mode as we were interviewing, you know, university students

and we, we sort of knew that the educators wanted some form of learning mode. They kept saying, right, where's your learning mode?

And so everyone was like, okay, now we have to build it. But it was really the students who I think really drove the point home for us because they of course used a different word, which is brain rot, but we heard them talking about brain rot and they, you know, realized that in the short term it could, they can use AI chat bots to like help 'em, you know, just finish an assignment, right?

But when it comes to actually like studying for their midterm and like understanding and internalizing the concepts, they wanted a version

of Claude where they didn't have to prompt it, you know, in all these different ways.

- Exactly. They, they didn't want to just, you know, give it an assignment, it just pops the answer, - Right? - And instead you give it an assignment, it guides you through the answers. If they're studying for a final can just show a flashcard

that helps them memorize and learn the content. So that is what learning mode is, is that just completely changing the interface in Claude.

So it it focused on learning. - And how long did this first version take to build? - So the initial version actually took

very short period of time. This is a number of people that was extremely passionate about, you know, adding this capability.

It took us about two weeks from start to finish and it was amazing, - Incredible.

- Yeah. - The other aspect of this of course is you know, we can work on these training programs, we can improve our products and our model, but then of course it's how do we partner with outside world? We're just, we're a tech company, we're a small part of this much broader ecosystem.

And so, so you've been doing a lot of the work we've done partnering with institutions like the teachers union, AFT would love

to hear more about what goes into those partnerships and why are we so focused on them. - Yeah, I mean you and me both but,

but yeah, I think again, something we're excited about is just like we have our classroom experience. Mine is relatively outdated. I was in the classroom before COVID,

like I know it's a very different- - Out there pre-COVID, pre-AI- Yeah, pre-COVID, pre-AI. It's like basically doesn't even matter anymore.

But we get to partner with these organizations to learn from teachers who are actually in the classroom and professors who are in universities

to understand the real problems that they're having in their schools and the real benefits like things that are going really well

and lean into both of those, whether it's with education materials to train teachers up or you know, product solutions

that give them more autonomy and tools. And so - Yeah. And the, the heart of this, this is like,

this is a collective issue, right? Like a humanity wide collective issue. And we are far from knowing every single thing that we need

to help resolve this. I think like the through line for all of our work is to bring more people into the conversation.

Like if enough people take AI fluency courses, our hope is that then they bring that knowledge to their institutions

and start these conversations. And to what Zoe was saying before, students are really smart and they're also really engaged in the desire to not have brain rot. Like some of the best feedback that we get comes from our student users who I think sometimes we don't give the credit for that we think, well they're gonna definitely wanna achieve with this. They're gonna, but it's an institutional problem, not necessarily a human, I guess motivation problem.

I think, I think that our, our best feedback, like from all of our product users indicates that they don't want this reliance on these models. They want to feel like their own human capabilities are augmented and improved by this collaboration with AI. So I'm like very proud of us in general in our products

for not optimizing for kind of the standard engagement metrics where we're not trying to optimize for retention

or you know, how much time you spend on the product or dependency on the product. And we make active product decisions, you know, now and into the future that sometimes actually encourage that greater augmented thinking

or encourage again times when you don't use AI or as the kids call it, touch grass, right? So I'm excited for us

to keep going down that pathway. - Yeah, and I actually, that was one of the most surprising things to me joining Anthropic was

that it's, it's not a growth optimized company like right? Like most SaaS companies wanna optimize for users retention,

all these things and has a much broader perspective on what success looks like in our products,

which I think is really interesting - In our product development. That's not just true for the education initiatives

that we had, but for everything else we build Yeah. Isn't about keeping users engaged in the product.

This really is about having AI be beneficially deployed and impact society.

- I think we touched on is that like every decision to use AI somewhere or not as a deliberate choice?

I don't think that we are kind of on a screen path towards AI and like being every single place

and hopefully the work that we do as a company and the things we make, the choices we make

to build our product in certain ways can lead by example and also invite people in to start realizing

that everything is a deliberate choice and it is just as good and sometimes better to just choose not to sometimes, right?

Like I would hate to see a future where teachers outsource to AI the parts that I think really make good education,

which is the connection pieces when you really understand your students and can spend time with them and AI can be used in so many ways that allow teachers to have more time to do that kind of work. And I'm excited for us to kind of over time talk

with institutions and discuss with them ways where we can amplify that knowledge

they already have, right?

The experts we partner with have generally a, a decently clear opinion on when AI is actively harming

the educational outcomes. And it's just our job to listen and to try to implement that either in our products

or in our educational program. - So we've talked a lot about, you know, what,

what our personal views are on AI education, what we're doing as a, as a company, but we definitely haven't solved it, so, so

what are the things we're still uncertain about? I mean I'd be curious for all your takes on this, you know, what are the things we're still trying to figure out.

- I have a couple things go ahead. Both pretty different, but one thing we touched on earlier is AI is changing, like

what it is that you need to teach. You brought up coding, we are pretty sure that coding curriculums will look very different in five years.

I'm interested to see how things start to shift and if there's any like frameworks or really anything that we can develop

to help academics along in these areas to understand what kinds of skills may be more augmented in the future

and which kinds of skills are gonna need additional human support like reviews or management.

I think we're starting to understand that in areas like computer science, but it's very, very early and we know that this is gonna affect a lot more fields.

Yeah. So especially in higher education, that's something I'm interested in. I hear a lot of concerns in K 12 about the different tools

and trying to understand what hap what happens to the data when you put it in those tools. And I think right now there's this like massive

proliferation of AI tools in classrooms and teachers and administrators are really overwhelmed for very good reasons.

Like there's a lot of concepts that are really new to everyone. There are elements of the data privacy that is new

and hard to understand. And so I'm really interested to see how that landscape evolves. Whether we need to really ramp up our education

around data privacy so people can better assess the landscape or whether we start to like, you know, see clear winners in this space.

I'll just be really interested to see what happens there. - I think in addition to that, I mean the technology is really changing rapidly.

So one of the concerns I'm not sure how it sort of will play out over time is how will institutions adapt?

They're generally slow moving and intentionally built that way. And the technology pace of change has been very rapid.

It's harder to predict like what would happen six months from now or a year from now. So I think that pace of change to

how institutions generally adapt a new technology is one area that I'm uncertain about. I - Also think that like just everywhere,

every institution feels an immense amount of pressure to just do something with AI instead of doing nothing.

And I, you know, I have no idea how to kind of balance or help organizations balance the fact

that pressure is really real, but also when it comes to education, especially moving

fast

and breaking things is not an option, right? And that's just so challenging for both the individual teacher but also the entire institution at large.

- I kind of call this unbundling of education. One is just the knowledge itself, which this AI is really good at

providing personalized education, but institutions provide more than just knowledge imparting knowledge into students.

The other is really like I have two children in, in college, it's not just learning they're getting theirs,

but also that's where they're growing, that's where they're maturing learning responsibility and and so forth. So what AI solves, you know,

very like really well is knowledge, impart knowledge and learning. I think what we would have to do

as society moving forward is how do we leverage AI in the knowledge transfer part but also retain these institutions for all

of the great roles that they play in society, - Right. To separate out some of the pieces such, exactly like the success metric of a good educator is not

to do every single part of this thing, but I think what you're saying is like yes, correct. Do more of one thing

and then let AI handle things that are like maybe knowledge like acquisition oriented but not like, I don't know

a relationship with the student, right? - Correct. Correct. I mean one of the feedback for example we've received when we were visiting

universities is that while AI assignments are very compelling, that's something they would like to do.

But AI assignment means students with AI therefore large assignment sets that might take say six months,

but they could end up taking like two weeks. How do you grade them? - Right? -

Right. So there is a lot of AI involvement means there's a lot of learning

that could happen much, much quickly as you leverage AI to do the learning aspect of it.

What about everything else that is those teachers institutions are providing the students. I think that is where that's unbundling

and you know, just leveraging all the right pieces from both the technology but what institutions do.

- Yeah. And I mean I think the best case scenario of this is reducing burnout at scale, which is like the number one issue

facing most of the teachers. - If you unbundle you can maybe do less? - Where every educator is really,

really talented at some things and those things bring them a lot of energy and they're exceptional at it. And what if AI could, you know, support them in the things

that don't bring them energy. And I think that just creates a much more well-rounded system for their personal lives and then also for the students that they're supporting.

- This conversation is also reminding me of like a, a part of the AI Fluency curriculum we have for educators

that I found super compelling, which is to have AI be so integrated, like into the assignments,

the experience and so on. And to instead start grading AI use and not grade the

outcomes as much.

Right, right. Going back to like how do you think about these long-term projects or how things differ in this AI world, having that kind of engagement is, is very different and AI forward in a way that I'm excited for more institutions to adopt. - I think you're right on like, you know, in our sort of conversation one of the things have come up is when you create, you're not necessarily just creating the final result, but you've created, how did students arrive to that result? How did they use the technology? What is the back and forth that also becomes part of the what learning is.

- One of our, one of our chief marketing folks here one time sat down with me and just said, you know, I think the true power of AI is the process. - Yes. - And I think that that's really the core of

what we're getting at, right? Yeah. - Right. - I'm also curious from like a model training perspective, what we can do to reconcile how, like I think I was having a, I had a really great conversation with someone who studied the philosophy of epistemics, right?

Like how do you know something is true? And they made a great point about how AI more easily than any other kind of intelligence that humans have encountered can be really confident, right? Or can can say things that sound so realistic.

And it usually, in a human being, it takes a lot of charisma and practice to get to be someone who can say that

and not say true things. But with AI, you're kind of encountering all the time and our human ability to discern what is true

and what isn't is based on how we discern in other humans, which may not be successful when you're applying to AI.

Right? And that's a whole different thing to unravel and figure out how you can, do you match the AI personality more to that area?

Or do you start teaching people of a new way of discerning truth? And I mean my hot take is I think

that like the latter one is a little more powerful because it's also a good inoculation against all forms

of like persuasive writing, persuasive thinking regardless of whether or not it's an AI or a human being.

But that critical thinking, going back to the fact that AI is just forcing us to reconcile things that already exist, I think that teaching

that critical thinking is really hard. Right. - Yeah. And this goes back to like child psychology, right? Like we have this whole generation of digital natives now who can very clearly identify a spam text when that's like really difficult for some folks. And like what is the AI native generation?

What does that look like? And what impacts does it have on on kids development?

Like there's just things we don't know yet. - Well, to close this out, because I think we could go on forever, I would love to hear from all of you as we think about five years out,

what's a success look like for teaching and learning? - Five years is a crazy kind of time to predict the AI world.

I guess I'll give you my hope, right? I don't know if I know what success looks like, but I do know what I hope for, which is

that in educational institutions we have like teachers have so much more time to

engage individually in the relationships
and the fostering portion of it. You know, going back to Ephraim's point, like maybe
it's not the knowledge acquisition part of things
that teachers participate in as much versus like synthesizing that knowledge into the
greater ecosystem of your life
and the world and understanding how any given individual can best be helped to
learn.

'cause we're still all unique individuals in this future and celebrating and
emphasizing that uniqueness is something
that I hope we can get to with education, - Right? I think five years is a very long
time,
but I think success to me by then will look like every person on the planet has a
personalized tutor ready for them at any time. And then also if we've made that
transition successfully,
our institutions will have survived and are playing the vital role they already play in
our society.

- Yeah. For me, I think it's going back to that critical thinking piece. I want every
person really, every student, every teacher
to have a shared vocabulary and cultural understanding around what it means to
use AI and learning.

I think just a lot more discernment and reflection on and just being intentional about
using AI.

- I know I feel like I go back to this like a broken record, but wouldn't it be great if
like every single student could
articulate when they want to use AI when they don't want to and why? Like that
kind of knowledge about your own habits
and how you think and how you learn best. That personalization of knowledge is so
exciting.

And I'm like, that's a shorthand heuristic, I guess for what success could look like. -
Or maybe on the other hand, like they don't have to make
that decision because technology, - The technology, yeah. The product - Is the
product itself as well adopted, meaning both of those-

- Together, definitely the product and the experience and the education all have to
go hand in hand. Exactly. - I think the thing I keep going back to, which
to me brings optimism. 'cause some days in this work there's a lot of pessimism of
jobs are changing

and we have no idea what my, you know, our jobs are - And a lot of personal
responsibility that we feel over any, you know, things that happen.

- But at the same time, I do see in a world where, you know, intelligence is
becoming abundant

and in some ways commoditized, I think that will, you know, no longer be our
defining trait as humans.

And that can be scary at some point. But I also think that's liberating because for
the last few hundred years I almost feel like

we've lost some of our humanity as we, you know, you have the industrial revolution
and we're able to do all these things,

but we're also like going into offices and doing tasks and defining our ourselves by
the work that we do.

And that may not be the thing that we can do best, you know, five years from now.

But there are so many things that a teacher does
that a doctor does that are truly human, that are not intelligence, you know?
And so I, I almost am excited for the things to sort of strip away. And so for us, and
I think our education systems at the core
of it, to really focus on what makes this human. - There's an Oxford professor that
said this,
this great quote I think about all the time, which is he says, I think the age of AI will
be the age
of asking good questions. And like that's something that doesn't necessarily come
from knowing a lot, right? It's just about being curious
and then also being a little discerning and skeptical about things you get in return,
which then yield better questions. And I think with AI in our pockets, the personal
tutors,
like the world of the question space has opened up immensely and way beyond
anything else we've ever
had in human history. And we just kind of have to steward people towards the
mindset that allows them
to ask good questions. - There's never been a better time to have a problem. -
There's never been a better time to have a problem.
- I think that's a perfect way to end. Well thank you all. Thanks for taking the time.